When Retrieval Hurts: A Critical Analysis of RAG in Medical Question Answering

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Abstract

Retrieval-Augmented Generation (RAG) has been widely adopted to enhance Large Language Models (LLMs) by incorporating external knowledge, yet its effectiveness in medical question answering remains underexplored. This study presents a systematic evaluation of RAG systems on the MedQA-USMLE dataset, revealing counterintuitive findings that challenge the "always-RAG" paradigm. We evaluated three RAG approaches (BM25, Dense Retrieval, and Hybrid) against a pure LLM baseline using Qwen3-8B-Instruct as the primary LLM, with a knowledge base of 125,847 medical textbook chunks from 18 authoritative sources. Surprisingly, all RAG methods underperformed the pure LLM baseline (60.69%), with BM25-RAG, Dense-RAG, and Hybrid-RAG achieving 59.43%, 60.14%, and 59.91% accuracy, respectively. Our analysis reveals that RAG introduces a dual effect: while correcting 5-6% of questions, it simultaneously misleads 6-7% of cases, resulting in a net negative impact of 7-16 questions. Critically, we demonstrate that retrieval confidence scores fail to predict RAG utility (AUC $\approx$ 0.53, Cohen's d = 0.06), indicating current retrieval systems cannot distinguish helpful from harmful contexts. Oracle analysis suggests a theoretical upper bound improvement of +6.05% if perfect RAG selection were achievable, highlighting the urgent need for selective RAG strategies. These findings challenge the assumption that external knowledge universally improves medical AI systems and emphasize the importance of context-aware retrieval mechanisms for clinical safety.

CCS Concepts

• **Information systems → Retrieval tasks and goals**.

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Keywords

Medical Question Answering, Retrieval-Augmented Generation, Large Language Models

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1 Introduction

Retrieval-Augmented Generation (RAG) has become a dominant paradigm for enhancing Large Language Models (LLMs) in knowledge-intensive tasks [1], particularly in high-stakes domains like medicine where factual accuracy is critical [14]. However, a fundamental question remains underexplored: does retrieval augmentation always help, or can it sometimes harm performance?

While most RAG research focuses on demonstrating benefits [7], retrieval inherently introduces noise—irrelevant, outdated, or conflicting information that may override correct parametric knowledge [15]. Systematic analysis of these negative effects remains scarce, and whether retrieval scores can reliably predict augmentation utility has not been rigorously examined.

Our empirical study on MedQA-USMLE reveals surprising findings: all RAG methods (BM25, Dense, Hybrid) underperform the pure LLM baseline, with retrieval harming 6-7% of questions while helping only 5-6%. Oracle analysis shows a modest upper bound of +6.05% even with perfect selection, and retrieval scores fail to distinguish helpful from harmful cases (AUC=0.53). This paper makes three contributions:

(1) **Systematic quantification of negative effects**: We demonstrate that RAG actively misleads 6-7% of medical questions where the LLM alone would answer correctly.
(2) **Oracle analysis and score failure**: We establish a +6.05% theoretical ceiling and show that retrieval confidence cannot predict RAG utility (AUC=0.53, Cohen's d=0.06).

(3) **Paradigm shift toward selective RAG**: We argue for moving from "always-RAG" to context-aware selective strategies for safer medical AI deployment.

Section 2–6 present related work, methodology, results, discussion, and conclusions.

2 Related Work

RAG and Adaptive Retrieval. RAG combines retrieval with generative models. Lewis et al. [1] first proposed the RAG framework that retrieves relevant documents and conditions generation on them. REALM [2] introduced retrieval augmentation during pre-training. Dense passage retrieval [3] improved retrieval quality using learned embeddings, with contrastive learning further enhancing unsupervised retrieval [4]. Recent surveys [5] provide comprehensive overviews of RAG architectures. More recently, Self-RAG [6] enables models to reflect on retrieved content and decide when to use it. Self-BioRAG [7] extends self-reflection mechanisms to biomedical domains. RAG² [8] introduces rationale-guided retrieval for medical QA. AdaptiveRAG [9] learns to adapt retrieval strategies based on question complexity. However, most work focuses on demonstrating benefits rather than systematic analysis of negative effects.

Medical QA and RAG Failures. Medical QA benchmarks include MedQA [10], PubMedQA [11], and MMLU [12], with MedRAG [13] establishing RAG-specific evaluation. Studies show retrieval quality doesn't always predict performance [15], and retrieved content can conflict with parametric knowledge [16–18]. Our work provides the first systematic quantification of RAG's negative impacts in medical QA.

3 Methodology

Our framework (Figure 1) compares a pure LLM baseline against three RAG variants, with Oracle analysis providing the theoretical upper bound.

3.1 Dataset and Knowledge Base

We use the MedQA-USMLE validation set [10] containing 1,272 four-option multiple-choice questions covering diverse clinical topics including diagnosis, treatment, pathophysiology, and clinical reasoning. Each question presents a clinical vignette followed by options (A–D). Following established medical QA evaluation protocols [13], the model is prompted to output a single letter (A, B, C, or D) representing the selected answer, enabling direct accuracy computation and fair comparison with prior work. The knowledge base comprises 125,847 pre-segmented chunks from the MedRAG Textbooks corpus [13], extracted from 18 authoritative medical textbooks.

3.2 Retrieval Methods

BM25: Standard Okapi BM25 with $k_1 = 1.5$, $b = 0.75$:

$$\text{BM25}(d, q) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{\text{tf}_{t,d} \cdot (k_1 + 1)}{\text{tf}_{t,d} + k_1 \cdot (1 - b + b \cdot \frac{|d|}{\text{avgdl}})} \quad (1)$$

Dense: MiniLM-L6-v2 (384-dim) with FAISS inner product over L2-normalized vectors (cosine similarity).

Hybrid: Reciprocal Rank Fusion [20] with $k = 60$ and equal weights:

$$\text{RRF}(d) = \sum_{r \in \{\text{BM25,Dense}\}} \frac{0.5}{60 + \text{rank}_r(d)} \quad (2)$$

All methods retrieve top-5 documents, following MedRAG protocol [13].

3.3 Language Model and Metrics

We use **Qwen3-8B-Instruct** (temperature 0.1) as primary LLM, with Qwen-Plus validation in Section 5.2.

Impact categories: Both Correct, Both Wrong, Positive Impact (LLM wrong → RAG correct), Negative Impact (LLM correct → RAG wrong). Net Impact = Positive − Negative.

Oracle Upper Bound: Maximum accuracy if perfect RAG selection were possible:

$$\text{Acc}_{\text{Oracle}} = \text{Acc}_{\text{LLM}} + \frac{|\text{Positive Impact}|}{N} \quad (3)$$

Retrieval Quality: ROC-AUC and Cohen's d for score discrimination.

4 Results

4.1 Overall Performance

Table 1 reveals a counterintuitive finding: all RAG methods underperform the pure LLM baseline (60.69%). RAG methods range from 59.43% to 60.14%, representing degradations of 0.55–1.26 percentage points. While McNemar tests yield p-values of 0.21–0.63 (not significant at α=0.05), the consistent negative direction across all three methods suggests a systematic pattern rather than random fluctuation.

This trend aligns with RAG² [8], which reported similar effects: Llama-3-8B-Instruct dropped from 57.7% to 55.3% with MedCPT retrieval, and GPT-3.5 (5-shot) decreased from 53.6% to 51.0%. The reproducibility across different retrieval methods, model architectures, and research groups indicates this phenomenon reflects a fundamental challenge in medical RAG rather than methodological artifacts.

Table 1: Overall Accuracy Comparison.

Method	Accuracy	95% CI	Correct	Total	Δ vs LLM
Pure LLM	**60.69%**	[58.01, 63.38]	772	1,272	–
BM25-RAG	59.43%	[56.74, 62.13]	756	1,272	-1.26%
Dense-RAG	60.14%	[57.45, 62.83]	765	1,272	-0.55%
Hybrid-RAG	59.91%	[57.21, 62.60]	762	1,272	-0.78%

4.2 Impact and Oracle Analysis

To understand the mechanisms behind overall performance degradation, we decomposed results into four mutually exclusive categories based on whether the pure LLM and RAG system each answered correctly or incorrectly. Table 2 presents this impact analysis.

The critical finding is the asymmetry between positive and negative impacts: negative impacts (6.37–6.84%) consistently exceed positive impacts (5.11–6.05%) across all retrieval methods, yielding

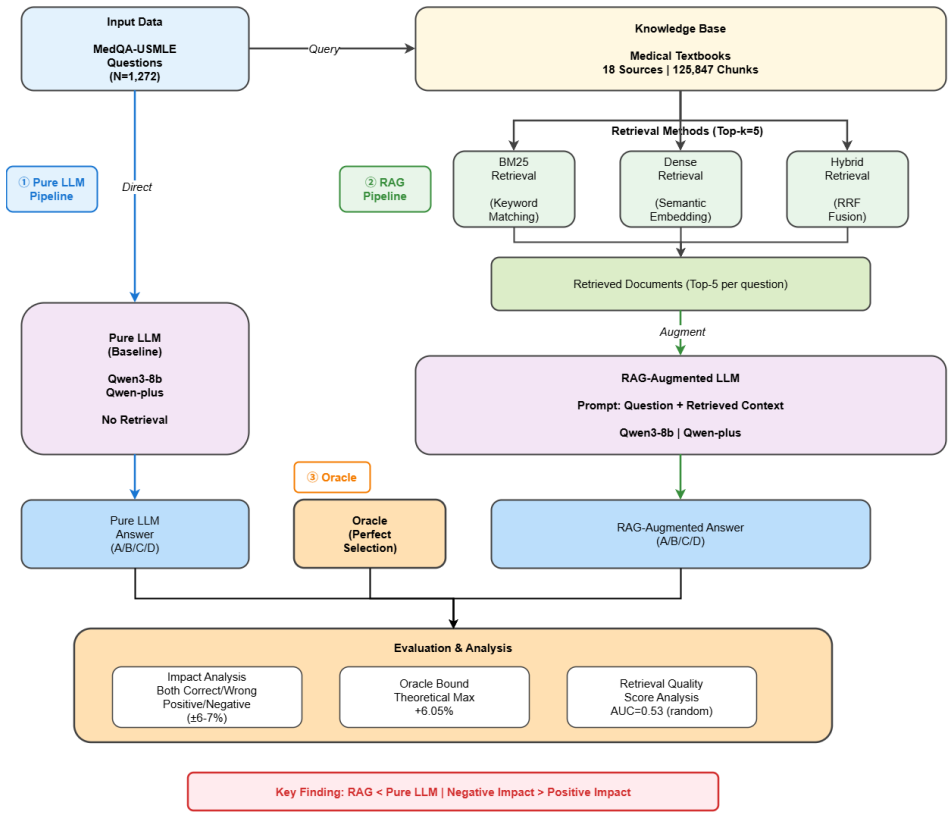


Figure 1: Evaluation framework comparing Pure LLM, three RAG methods (BM25, Dense, Hybrid), and Oracle selector on MedQA-USMLE.

Table 2: Detailed Impact Analysis.

Method	Both Correct	Both Wrong	Negative	Positive	Net
BM25-RAG	691 (54.3%)	435 (34.2%)	81 (6.37%)	65 (5.11%)	**-16**
Dense-RAG	690 (54.2%)	425 (33.4%)	82 (6.45%)	75 (5.90%)	**-7**
Hybrid-RAG	685 (53.9%)	423 (33.3%)	87 (6.84%)	77 (6.05%)	**-10**

net losses of 7–16 questions. For approximately 87–88% of cases, retrieval makes no difference—both systems answer correctly or both fail. However, among cases where RAG changes the outcome, it harms more often than it helps.

Figure 2 visualizes this asymmetry: the left panel shows negative impacts (red) consistently exceed positive impacts (blue) across all RAG methods. BM25-RAG shows the worst net effect (-16), while Dense-RAG performs relatively better (-7) due to higher positive impact counts. The right panel presents the Oracle upper bound analysis, which assumes perfect knowledge of when retrieval will be beneficial. Even with such ideal selective application, the maximum achievable accuracy is only 66.74%, representing a modest +6.05% improvement over the 60.69% baseline. This limited theoretical ceiling indicates that under our experimental setup, selective gating strategies alone offer constrained gains—substantial improvements likely require fundamental advances in retrieval quality and knowledge integration.

4.3 Retrieval Score Analysis

A critical question for practical RAG deployment is whether retrieval confidence scores can predict when augmentation will be beneficial versus harmful. If such prediction were reliable, we could implement selective RAG strategies that apply augmentation only when retrieval quality exceeds certain thresholds. We investigated this using Hybrid-RAG, comparing RRF scores between 87 negative impact cases and 77 positive impact cases.

Table 3: Retrieval Score Statistics.

Group	Mean	Std Dev	N	Cohen's d [95% CI]
Negative Impact	330.48	122.61	87	0.06 [-0.27, 0.38]
Positive Impact	337.44	120.01	77	
Tests: t-test p=0.443; ROC-AUC=0.53				

Table 3 reveals that score distributions are nearly identical: the mean difference is only 6.96 points (337.44 vs 330.48), with Cohen's d=0.06 indicating a negligible effect size. The t-test yields p=0.443, confirming no significant difference.

Figure 3 provides comprehensive visualization of this finding. Panel (a) shows the score distributions for positive and negative impact cases, which overlap almost completely with nearly identical means. Panel (b) presents the ROC curve with AUC=0.53, barely



Figure 2: Impact breakdown (left) and Oracle upper bound (right). Negative impacts exceed positive impacts; Oracle's maximum accuracy is 66.74% (+6.05%).

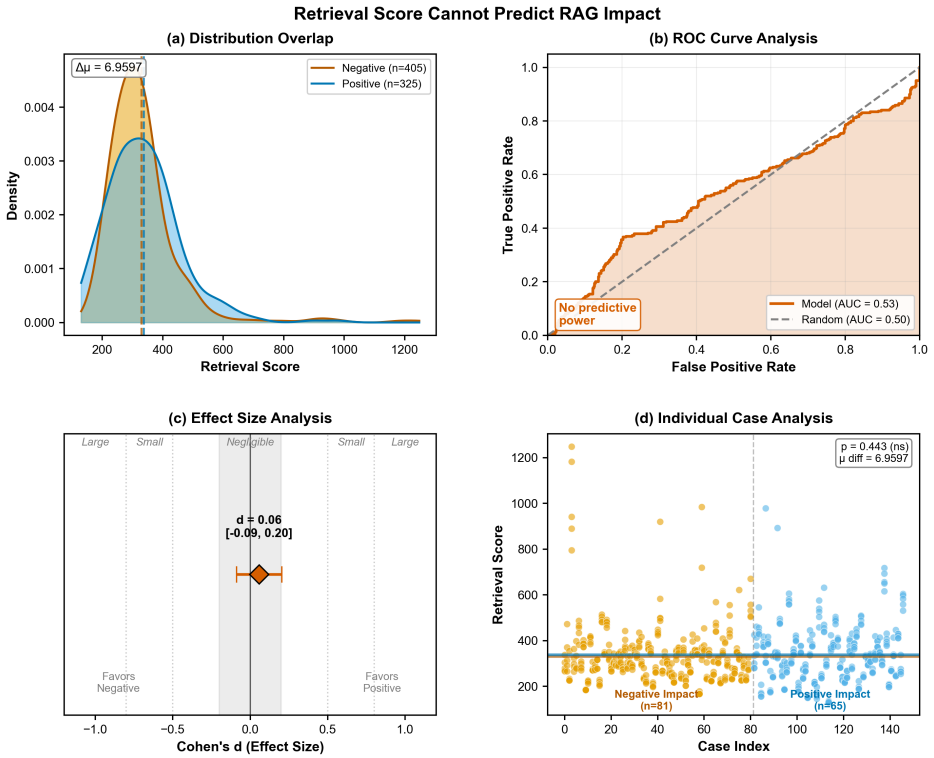


Figure 3: Retrieval scores fail to predict RAG utility. (a) Distribution overlap: nearly identical score distributions for positive and negative impact cases. (b) ROC curve: AUC=0.53 indicates chance-level discrimination. (c) Effect size: Cohen's d=0.06, well within the "negligible" range. (d) Individual case analysis showing complete overlap between groups.

above the 0.50 random baseline, demonstrating that retrieval scores have essentially no discriminative power. Panel (c) displays the effect size analysis, with Cohen's d=0.06 falling squarely in the "negligible" zone. Panel (d) shows individual case scores, confirming the complete overlap between harmful and helpful retrievals.

This finding has profound implications: current RAG systems cannot use retrieval confidence as a quality signal. The assumption that higher-scoring retrievals are more trustworthy and more likely to improve performance is empirically unfounded in the medical QA context.

5 Discussion

5.1 Root Causes of RAG Performance Degradation

Manual analysis of 81 negative cases reveals three failure types: **context mismatch** (35%)—retrieval finds relevant terms but wrong clinical context; **phrase-level extraction** (40%)—models extract phrases without broader understanding; **knowledge fragmentation** (25%)—when retrieved chunks contain partial or incomplete information that, while individually accurate, creates logical inconsistencies when combined (e.g., one chunk states "Drug A inhibits

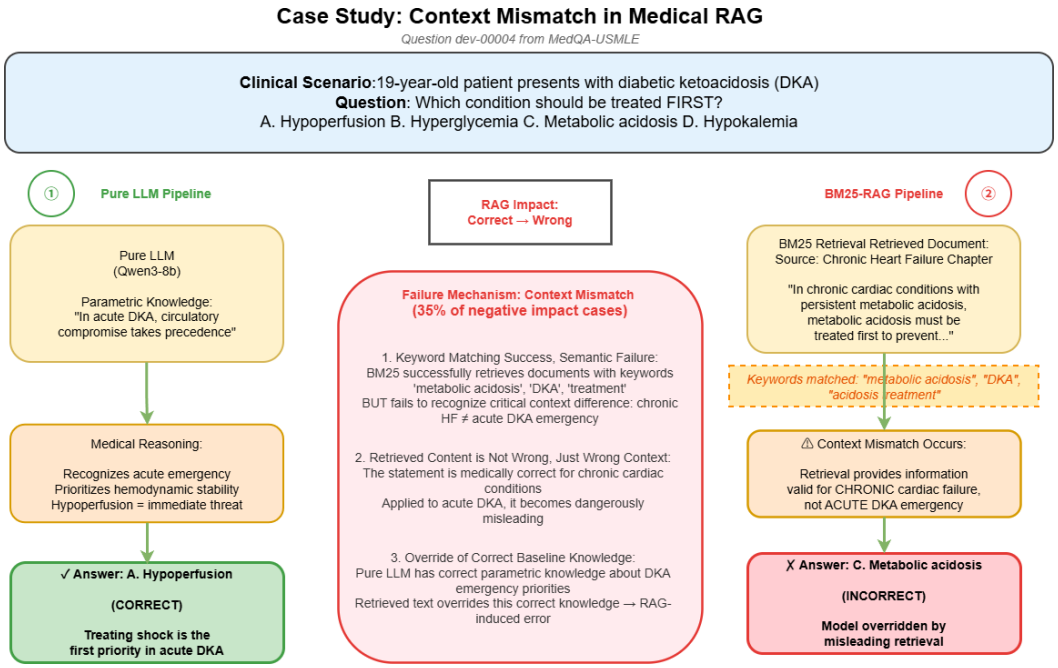


Figure 4: Context mismatch: DKA case where retrieved chronic guidance causes incorrect answer.

CYP3A4" while another states "Drug B is metabolized by CYP3A4," but neither explicitly mentions the interaction consequence, causing the model to fail in synthesizing a coherent clinical conclusion). Figure 4 illustrates context mismatch: a DKA emergency case where retrieved chronic heart failure guidance ("treat acidosis first") overrode the LLM's correct prioritization of hypoperfusion.

5.2 Dual Nature and Safety Implications

RAG exhibits a dual nature: it helps 5–6% of questions by filling knowledge gaps but harms 6–7% by overriding correct parametric knowledge, yielding net negative impact. Critically, these negative cases represent instances where the pure LLM would have answered correctly without retrieval. Validation with **Qwen-Plus** [19] (85% baseline → 83% with RAG) confirms this effect generalizes across model scales and is not limited to open-source models. For Qwen-Plus, we re-run the same MedQA-USMLE validation set (1,272 questions) with identical prompts and retrieval settings.

The 6–7% negative impact rate raises serious concerns for clinical deployment. While this percentage may appear modest, its significance must be evaluated in the context of medical practice where even small error rates can affect patient safety and clinical decision-making. In high-volume clinical AI applications, this rate would translate to a substantial number of potentially harmful recommendations.

6 Conclusion

This study provides systematic evidence that RAG can harm medical QA performance. All three RAG methods (BM25, Dense, Hybrid) underperform the pure LLM baseline on MedQA-USMLE, with 5–6% positive impacts outweighed by 6–7% negative impacts. Oracle analysis shows only +6.05% maximum improvement with perfect selection, and retrieval scores fail to predict utility (AUC=0.53).

Our findings demonstrate that optimizing *when* to apply RAG is insufficient—the modest Oracle ceiling suggests fundamental improvements in retrieval quality and knowledge integration are needed. The failure of confidence scores to distinguish helpful from harmful cases undermines standard quality control assumptions and raises safety concerns for clinical deployment. These results hold under our MedQA-USMLE validation setting and textbook-based knowledge corpus; generalization to other tasks or corpora may vary.

Limitations. Our study has four limitations: (1) single dataset (MedQA-USMLE); (2) single knowledge base (textbooks only); (3) primary model Qwen3-8B-Instruct, with limited Qwen-Plus validation; (4) single-run evaluation without multiple seeds.

We advocate shifting from "always-RAG" to selective, context-aware strategies. Future work should focus on: (1) developing reliable utility prediction mechanisms beyond retrieval scores; (2) designing context-aware retrieval that recognizes clinical nuances and acute vs. chronic distinctions; (3) implementing robust quality control that can identify potentially harmful retrievals before they affect outputs; and (4) exploring hybrid approaches that leverage RAG's benefits while mitigating its risks. In safety-critical medical AI, patient safety must take precedence over performance metrics.

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References

[1] P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[2] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M. Chang, "REALM: Retrieval-Augmented Language Model Pre-Training," in *Proceedings of ICML*, 2020.

[3] V. Karpukhin, B. Oğuz, S. Min, et al., "Dense Passage Retrieval for Open-Domain Question Answering," in *Proceedings of EMNLP*, 2020, pp. 6769–6781.

[4] G. Izacard, M. Caron, L. Hosseini, et al., "Unsupervised Dense Information Retrieval with Contrastive Learning," *Transactions on Machine Learning Research*, 2022.

[5] Y. Gao, Y. Xiong, X. Gao, et al., "Retrieval-Augmented Generation for Large Language Models: A Survey," *arXiv preprint arXiv:2312.10997*, 2023.

[6] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection," in *Proceedings of ICLR*, 2024.

[7] M. Jeong, et al., "Improving medical reasoning through retrieval and self-reflection with retrieval-augmented large language models," *Bioinformatics*, 40(Suppl_1): i119–i129, 2024.

[8] M. Su, H. Zhou, P. Shi, et al., "Rationale-Guided Retrieval Augmented Generation for Medical Question Answering," *arXiv preprint arXiv:2411.00300*, 2024.

[9] J. Jeong, B. Choi, S. Hwang, et al., "Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity," *arXiv preprint arXiv:2403.14403*, 2024.

[10] D. Jin, E. Pan, N. Oufattole, et al., "What Disease does this Patient Have? A Large-scale Open Domain Question Answering Dataset from Medical Exams," *Applied Sciences*, vol. 11, no. 14, p. 6421, 2021.

[11] Q. Jin, B. Dhingra, Z. Liu, W. Cohen, and X. Lu, "PubMedQA: A Dataset for Biomedical Research Question Answering," in *Proceedings of EMNLP-IJCNLP*, 2019, pp. 2567–2577.

[12] D. Hendrycks, C. Burns, S. Basart, et al., "Measuring Massive Multitask Language Understanding," *arXiv preprint arXiv:2009.03300*, 2020.

[13] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking Retrieval-Augmented Generation for Medicine," *arXiv preprint arXiv:2402.13178*, 2024.

[14] K. Singhal, S. Azizi, T. Tu, et al., "Large Language Models Encode Clinical Knowledge," *Nature*, vol. 620, pp. 172–180, 2023.

[15] F. Cuconasu, G. Trappolini, F. Siciliano, et al., "The Power of Noise: Redefining Retrieval for RAG Systems," *arXiv preprint arXiv:2404.15488*, 2024.

[16] Y. Xie, K. Kawaguchi, Y. Zhao, et al., "Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts," *arXiv preprint arXiv:2305.13300*, 2023.

[17] S. Longpre, K. Perisetla, A. Chen, et al., "Entity-Based Knowledge Conflicts in Question Answering," in *Proceedings of EMNLP*, 2021, pp. 7052–7063.

[18] H. Chen, H. Liu, C. Chen, et al., "Rich Knowledge Sources Bring Complex Knowledge Conflicts: Recalibrating Models to Reflect Conflicting Evidence," in *Proceedings of EMNLP*, 2022, pp. 2292–2307.

[19] J. Achiam, S. Adler, S. Agarwal, et al., "GPT-4 Technical Report," *arXiv preprint arXiv:2303.08774*, 2023.

[20] G. V. Cormack, C. L. Clarke, and S. Buettcher, "Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods," in *Proceedings of SIGIR*, 2009, pp. 758–759.